

Co-Curricular Activity

FPGA Development Board

Bi-Weekly Progress Report – 01

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1 Project Objective

The objective of the project relies on supporting the operation of the ECP5 FPGA device by providing a complete development platform for it. Our current objective is to identify the requirements of the ECP5 module and to design a system level architecture of what and how we are supposed to achieve that. The project will be developed in a modular manner with each subsystem being designed independently and then integrated into a complete system.

2 Initial Understanding of the Problem

The first stage of the problem is to identify what are the major subsystems of an FPGA development board and what are the requirements of each subsystem as well as what does it provide to the overall system. The major subsystems of an FPGA development board that we identified are as follows:

1. FPGA core and associated power supplies
2. FPGA configuration and programming interface
3. Clocking and timing circuitry
4. User I/O and peripheral interfaces
5. Debugging and monitoring facilities
6. PCB layout and mechanical considerations

The exact implementation of these subsystems will depend on the selected ECP5 device/package, required peripherals, available board area, and project requirements.

3 Work Undertaken: Power Subsystem

Worked on by Pranjali Pandey

The investigation of power subsystem requirements is ongoing and will be used to make informed decisions about the regulator selection and overall power architecture. Though the system is dependent on the requirements provided by the sub systems, I have made a preliminary design investigation based on the Lattice ECP5 evaluation board and the documentation provided.

From this preliminary investigation, power requirements have been identified for the following domains:

- Required FPGA supply domains
- Operating voltage ranges
- I/O bank supply requirements
- Configuration-related supply requirements
- Current requirements and power considerations
- Power-up and supply sequencing requirements

At this stage the purpose of the investigation is not to immediately select regulator components, but to identify the general power end points and work backwards towards the architecture of the power subsystem.

3.1 Identification of Required Power Domains

The preliminary power domains identified from the ECP5 documentation include the following categories:

Domain	Purpose
VCC	FPGA core supply
VCCAUX	Auxiliary supply
VCCIOx	I/O bank supply
Configuration supply	Configuration/JTAG related
Peripheral rails	External peripherals

Table 1: Preliminary power-domain identification

The exact number of VCCIO rails required will depend on the selected FPGA package and the allocation of I/O banks to different interfaces.

This makes FPGA package selection and I/O planning an important dependency for the final power architecture.

The external peripheral rails will depend on the specific peripherals and thus will have to wait until the peripheral selection is complete.

3.2 Reference Design Study

Lattice’s ECP5 evaluation board documentation has been used as a reference for understanding how a practical ECP5 development platform can be constructed.

The purpose of studying the evaluation board is not to reproduce the design directly. Instead, it provides a practical reference for identifying the types of supporting circuits and power domains that may be required.

The evaluation board architecture is being used as a starting point for determining:

- Power supply architecture
- FPGA configuration method
- JTAG programming interface
- SPI Flash configuration
- Clocking arrangements
- User I/O
- Debugging facilities

The final board architecture will be adapted according to the requirements and constraints of the present project.

3.3 Preliminary Power Architecture

The current design approach is to define the required power endpoints first and then work backwards to determine how these supplies should be generated.

The preliminary chain is:

Input Power → Power Regulation → Required Voltage Rails → FPGA / I/O Banks /
Peripherals

For each required rail, the following parameters will be determined:

1. Required voltage
2. Estimated current requirement
3. Expected peak/transient requirement
4. Source/input voltage
5. Regulator topology
6. Decoupling requirements
7. Monitoring and test points

Once these requirements are established, suitable regulator and supporting components can be selected and incorporated into the BOM.

4 Conclusion

The first stage of the project has been focused on converting the general requirement of developing an ECP5-based FPGA board into a set of identifiable engineering problems.

The power subsystem has been selected as the initial area of detailed work. The current approach is to identify all required power endpoints, determine their electrical requirements, and subsequently work backwards toward regulator selection and a complete power architecture.

Further work will integrate this subsystem with FPGA configuration, clocking, I/O, debugging and PCB layout to produce the complete development-board design.